

NYAS FALL 2023 PROJECT PROPOSAL

PETransformers: Resourcing Polyethylene Using Biochemistry

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The Problem

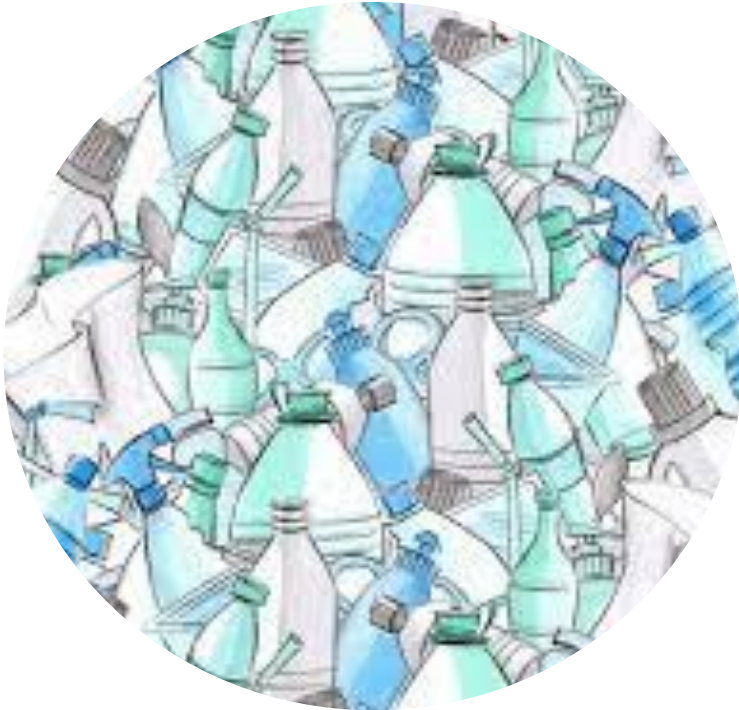
The problem we attempted to solve and
relevant background information

The Environmental Crisis of Plastic

As we navigate in our daily lives, the abundance of plastic tends to be prevalent. The planet that was once thriving is now suffocating under the weight of plastics, with habitats destroyed, oceans drowned with plastic, and wildlife at the brink of endangerment. Behind all of this are ascending greenhouse gas emissions, and the growing health concerns, casting an layer of uncertainty over our future.



Background



Polyethylene terephthalate (PET) is the most commonly utilized plastic, commonly found in containers, fibers, and agriculture due to its durability, affordability, and convenient processability (Cui et al., 2021).

Prior research indicates that modified *E. coli* can function as a whole-cell biocatalyst for the biodegradation of PET (Qi et al., 2021).

Recycled PET from a plastic bottle has the potential to be transformed into vanillin through the integration of the pathway with enzyme-catalyzed PET hydrolysis (Sadler & Wallace, 2021).



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Our Hypothesis

Our general prediction for our desired
outcome that we are testing

Hypothesis

Our hypothesis:

If we use PETase and MHETase enzymes on polyethylene and E. Coli and alcohol dehydrogenase on the products terephthalic acid and ethylene glycol respectively, then we will obtain glycolic acid and vanillin. This is based off of previous studies on the individual portions of the process. This process will work to eliminate large amounts of plastic in landfills and turn them into something more beneficial to society (vanilla flavouring and skincare products). Using the two processes as described before, we can tackle the whole process of polyethylene -> vanillin + glycolic acid.

What we changed:

- ★ Realizing the difficulty of the ethylene glycol breakdown and also that ethylene glycol is not environmentally harmful, we decided to leave it as is and use it in the industry as itself rather than glycolic acid.
- ★ We decided not to use alcohol dehydrogenase because we believed it to be a bit too ambitious for our time constraints.
- ★ To collect the vanillin released by the E.coli after the degradation of vanillin, we designed a 2-chamber containment system using dialysis tubing



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Methods

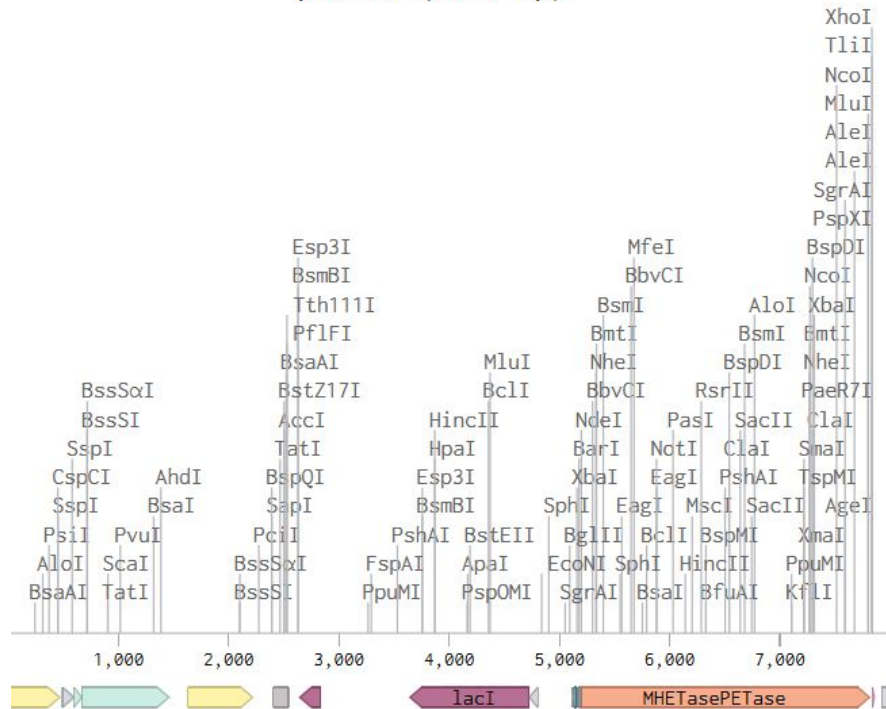
How we tested our hypothesis

Benchling Process

3 plasmids were designed:

- 1 pCJ190 plasmid containing a MHETase-PETase gene insert to break down PET
- 2 pUC19 plasmids to transform TPA to vanillin
 - Plasmid 1: TphA1, TphA2, and TphA3 gene inserts
 - Plasmid 2: DCDDH gene insert. Used for analyzing the sequences that are involved in the process in converting PET to vanillin and mapping out the genetic constructs

pCJ190 (7997 bp)

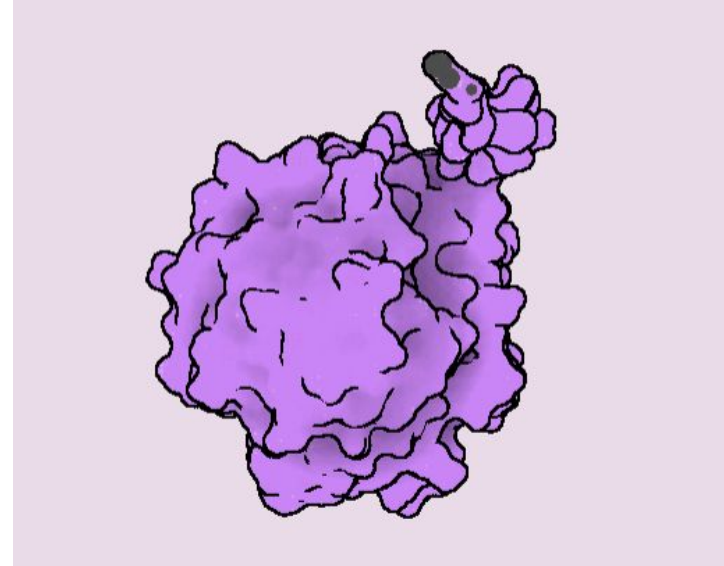


Linear Plasmid Map for pCJ190

AlphaFold

Protein Design

Using AlphaFold software, we computationally designed each of the proteins that the E.coli produced throughout our multi-step chemical process. This included 4 proteins: Carboxylic acid reductase, Catechol o-methyltransferase, 1,2-dihydroxy-3,5-cyclohexadiene-1,4-dicarboxylate dehydrogenase and Putative terephthalate 1,2-dioxygenase



We also designed Leaf Branch Compost Cutinase (LCC). The LCC functions in breaking down plastic into the basic molecules/components that are a lot easier to recycle and make use of instead of leaving it to harm the environment.

Survey

We distributed 2 surveys, 1 with verified scales and 1 with community interest questions.

We distributed the survey throughout our schools, social media, Launchpad, etc.

Analyzed the results using data analysis for the verified survey; for the other we used it as a way to see what we will need to do to make our solution work better (i.e. more education, more awareness, more purification processes, etc)

10/18. How much PET (a type of plastic) waste do you think is produced globally each year? Do not search anything up, this is just your estimate.

0-20 million metric tons

20-40 million metric tons

40-60 million metric tons

60-80 million metric tons

80-100 million metric tons

100+ million metric tons

Prefer not to answer

Sample question from our community interest survey



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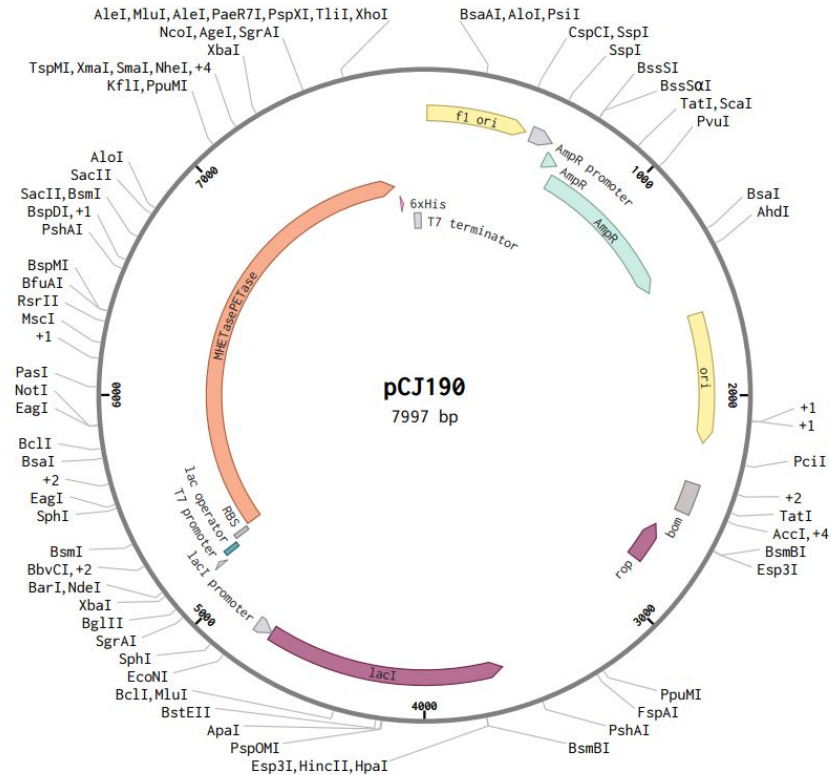
Results

What we learned from methods and how it
fits into the hypothesis

Bacteria 1: Plasmid Design

To convert PET (poly(ethylene) terephthalate) into terephthalic acid (TPA) and ethylene glycol (EG). A plasmid map of pCJ190, displaying a plasmid with an MHETase-PETase gene insert, is shown to the right. The C-terminus of MHETase has been linked to the N-terminus of PETase and has a His-tag at the C-terminus of PETase. This plasmid will be expressed in E. Coli BL21 due to their high protein expression rates.

Sourced from work done by Dr. Gregg Beckham and associates

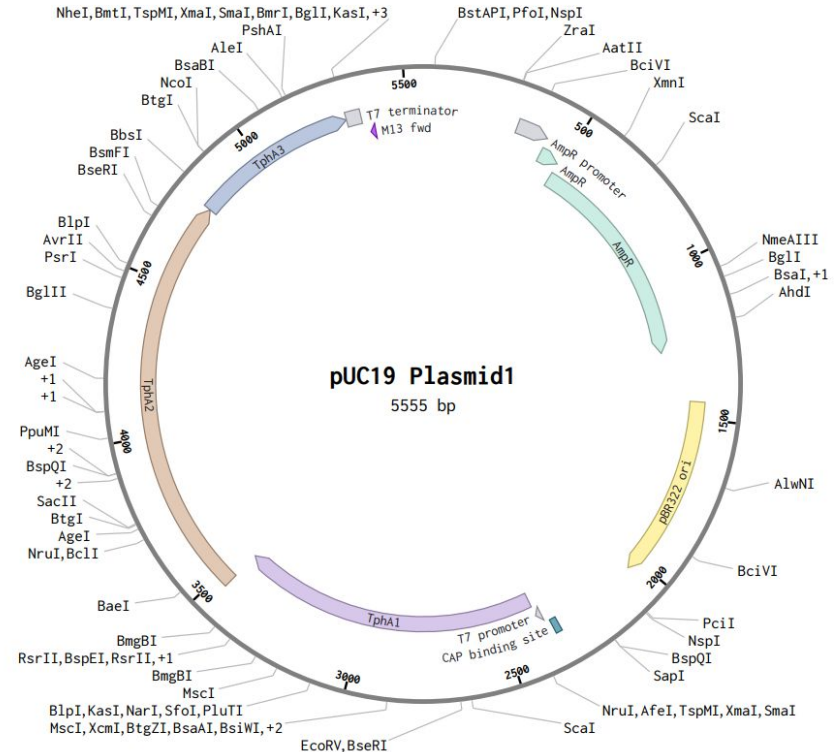


Bacteria 2A: Plasmid Design

To synthesize TPA into vanillin, two pUC19 plasmids are being used.

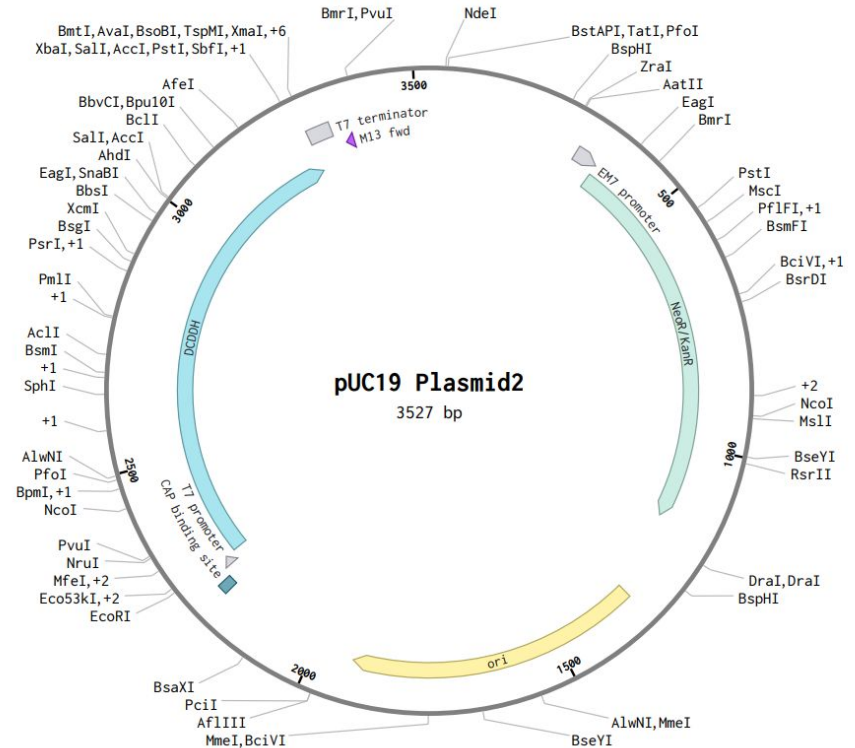
The first with TphA1, TphA2 and TphA3 inserts, which serve as a catalyst for initial structural modifications.

Similarly, *E. Coli* BL21 is used for high protein secretion.



Bacteria 2B: Plasmid Design

The second with DCDDH for later stages in synthesizing vanillin through biochemical reactions, transforming the modified TPA into our final product.

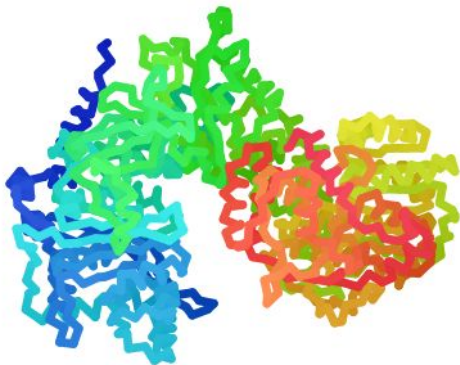


AlphaFold

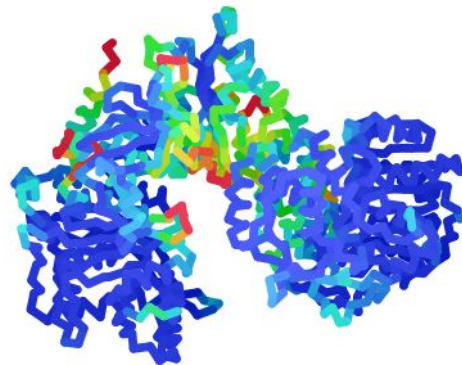
Colored protein models

Using the gene sequences of each protein we got from the UniProt database, we used AlphaFold to generate models and collect data for us on each of their structures

colored by N→C



colored by pLDDT



These are colored models of Carboxylic Acid Reductase (released by the E.coli), which catalyzes the reduction of a very wide range of aliphatic carboxylic acids as well as many aryl carboxylic acids.

Survey Data

Data from the survey conducted informed us that respondents aged 12-17 had the most reservations against using our product because it seemed unnatural and processed and because they had little knowledge of what it actually was and how it could affect their health. The presence of E. coli in the production process appears to raise red flags amongst consumers, affecting their attitudes, interest in green products, and willingness to engage in environmental action. On the other hand, products produced through methods not involving E. coli generated more positive responses, fostering favorable attitudes and encouraging both interest and proactive participation in environmental causes. This highlights the significance of eco-friendly production practices in shaping consumer behavior and promoting a greener mindset amongst the public.

How willing would you be to consume bacterially synthesized vanillin? 31 ⓘ

15/18 - How willing would you be to consume bacterially synthesized vanillin?	Percentage
0 (Never)	3%
1	3%
2	19%
3	35%
4	10%
5 (Very likely)	23%
Prefer not to say	6%

Results from "How willing would you be to consume bacterially synthesized vanillin?"



05 Solution

Where we can go from here, and where
we'll improve

Our Research and Discussion

Containment System

To collect all the vanillin released from the bacteria, we designed a containment system using 2 chambers of dialysis tubing as it has a pore size smaller than 1 micron, allowing TPA to move through the system while keeping bacteria stationary (to prevent plasmid conjugation).

Protein Mutations

For further research, we were considering mutating the proteins and plasmids used by the E.coli to generate a higher yield of vanillin in the end but, this would require complex lab equipment to actually be tested

Impact on a Global Scale

Our ever increasing reliance on plastics and plastic products is killing our Earth, and we need a way to stop it. Recycling has been lauded by eco-friendly corporations and nonprofits for its beneficial aspects, but why not go one step further? By resourcing zero-value waste into a valued product, we can not only help with the intense issue of plastic in landfills plaguing much of our world today, we can also increase consumers' access to environmentally friendly products.

Watch our Video!



Bibliography

Sources

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- Qi, X., Yan, W., Cao, Z., Ding, M., & Yuan, Y. (2021). Current advances in the biodegradation and bioconversion of Polyethylene Terephthalate. *Microorganisms*, 10(1), 39. <https://doi.org/10.3390/microorganisms10010039>
- Sadler, J. C., & Wallace, S. (2021). Microbial synthesis of vanillin from waste poly(ethylene terephthalate). *Green Chemistry*, 23(13), 4665–4672. <https://doi.org/10.1039/d1gc00931a>

Others

- Beckham, G. (n.d.). *PCJ190 (plasmid #162667)*. Addgene. [Addgene: Gregg Beckham Lab Materials](#)

